PROJECT REQUIREMENTS SPECIFICATION

Baseline requirements and scope definition

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

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Document Control

Field	Value
Document	Project Requirements Specification (PRS)
Project	AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)
Version	1.0 — Baseline Draft
Prepared for	SCTCE B.Tech CSE Project
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Status	Requirements baseline for design and implementation

1. Purpose

This document establishes the agreed functional scope, actors, system boundaries, constraints, assumptions, priorities, and acceptance expectations for the proposed multimodal assistive application. It is intended to be frozen before detailed implementation.

2. Project Vision

The system is a unified assistive platform intended to support people with Mild Cognitive Impairment through memory assistance, routine management, familiar-person recognition, object-location assistance, contextual conversation retrieval, safety monitoring, and caregiver support.

The project materials describe the motivation as a need to reduce dependence on multiple disconnected applications and to provide a privacy-preserving multimodal companion. The supplied project material also proposes smartphone capture of audio, video and GPS with heavier AI processing handled by a secondary/local node.

3. Scope

3.1 In Scope

- Patient mobile application with a simple, accessible interface.
- Caregiver interface/dashboard for patient management and monitoring.
- Object detection, tracking and last-observed object-location retrieval.
- Facial detection and recognition of registered/familiar individuals.
- Contextual conversational memory using speech-to-text, embeddings, retrieval and an LLM/RAG layer.
- Personal AI assistant capable of accessing structured application data and retrieved memories.
- Daily schedule, calendar/appointments and reminder management.
- Medication schedule and medication reminders.
- GPS-based geofencing and caregiver alerts.
- Relaxation, meditation and therapeutic/well-being audio features.
- Notifications and activity/event logging.
- Privacy, authentication, authorization and controlled access to sensitive data.

3.2 Out of Scope for the Baseline

- Medical diagnosis or clinical decision-making.
- Automatic prescription, dosage modification or medication changes by AI.
- Claiming physical ingestion of medication unless an actual verification mechanism is implemented.
- Guaranteed current physical location of an object; the system reports a last detected/observed location.
- Identity claims for unregistered people when recognition confidence is insufficient.
- Predictive health analytics, smart-home integration, wearables/IoT integration and secure cloud synchronization as mandatory baseline features; these remain future-scope items.

4. Stakeholders and Actors

Actor	Role	Primary Needs
Patient	Primary end user	Simple assistance, reminders, memory cues, object finding, familiar-person recognition, schedule and conversational support.
Caregiver	Remote support and configuration	Patient status, schedules, medications, safe zones, alerts, familiar-person registration and activity information.
AI/Backend System	Processing layer	Receive multimodal input, run AI services, retrieve data, generate responses and trigger actions.
Notification Service	Delivery mechanism	Deliver reminders, alerts and emergency notifications.

5. Functional Requirement Baseline

ID	Requirement	Priority	Acceptance / Verification
FR-001	The system shall authenticate patients and caregivers before access to protected functionality.	High	Valid credentials create an authenticated session; invalid credentials are rejected.
FR-002	The system shall maintain separate patient and caregiver access privileges.	High	A user cannot access functions outside their assigned role.
FR-003	The patient shall be able to view a daily schedule.	High	Today's scheduled activities are displayed in chronological order.
FR-004	The caregiver shall be able to create, edit and delete patient schedule items.	High	Changes are persisted and visible to the patient.
FR-005	The system shall generate reminders for configured activities, appointments and medications.	High	A reminder is generated at its configured time.
FR-006	The caregiver shall be able to create medication records and medication schedules.	High	Medication name, dosage/instructions and schedule can be saved and edited.
FR-007	The patient shall be able to ask the assistant about schedules, reminders and stored information.	High	Assistant retrieves structured data and returns a response.
FR-008	The system shall capture camera input for object and face processing.	High	Camera permission and capture workflow operate on supported devices.

FR-009	The system shall detect configured everyday objects from supported camera input.	High	A supported object is detected with a confidence score above the configured threshold.
FR-010	The system shall store object observations with time and contextual/approximate location information.	High	Observation records can be retrieved after a detection.
FR-011	The patient shall be able to ask where an object was last detected.	High	System returns the latest valid observation and timestamp.
FR-012	The system shall recognize registered/familiar people using facial features/embeddings.	High	Registered person is returned when confidence exceeds threshold.
FR-013	The caregiver shall be able to register familiar people and relationship information.	High	Person profile and recognition data can be stored and updated.
FR-014	The system shall identify an unknown person as unknown when no reliable match exists.	High	Low-confidence matches do not produce a false identity.
FR-015	The system shall convert supported speech input to text for conversational assistance.	Medium	Speech is transcribed into usable text where supported.
FR-016	The system shall store selected conversations/events as searchable memory representations.	High	Eligible records can be embedded/indexed for retrieval.
FR-017	The RAG module shall retrieve relevant memories for a user query.	High	Top relevant records are returned before response generation.
FR-018	The system shall use recognized-person context, where available, to improve memory retrieval.	High	Queries involving a known person can be filtered/weighted using identity context.
FR-019	The assistant shall generate responses grounded in available structured data and retrieved memories.	High	Response is traceable to available application context.
FR-020	The system shall allow caregivers to define geofenced safe zones.	High	A safe zone with center/radius can be created and modified.
FR-021	The system shall monitor patient location for configured geofence rules when location permission/service is available.	High	Location events are evaluated against the active safe zone.
FR-022	The system shall generate a caregiver alert when a configured geofence violation occurs.	High	Caregiver receives an alert with relevant time/location information.
FR-023	The system shall provide relaxation/meditation/well-being audio content.	Medium	Patient can start and stop supported sessions.
FR-024	The system shall maintain activity/event logs required for patient and caregiver views.	Medium	Relevant events have timestamp and event type.

FR-025	The system shall provide caregiver notifications for configured important events.	High	Configured alert conditions produce notifications.

6. Non-Functional Requirements

ID	Requirement	Priority	Acceptance / Verification
NFR-001	Privacy	High	Sensitive audio, video, face, conversation and location data shall be protected and minimized.
NFR-002	Security	High	Authenticated access, role-based authorization and secure transport/storage shall be used.
NFR-003	Usability	High	Patient interface shall use simple navigation, clear labels, large touch targets and optional voice interaction.
NFR-004	Performance	High	Normal interactions should return within a practical interactive delay; AI latency shall be measured during testing.
NFR-005	Reliability	High	Failure of one non-critical AI module shall not crash the entire application.
NFR-006	Maintainability	Medium	AI modules and backend services shall be modular and independently testable.
NFR-007	Scalability	Medium	The architecture should allow additional patients and modules without redesigning the entire system.
NFR-008	Auditability	Medium	Important configuration changes and alerts should be timestamped and logged.
NFR-009	Safety	High	The system shall clearly distinguish last-known/last-detected information from guaranteed current state.

7. Data Requirements

Entity	Purpose	Key Data
User	Authentication and role	user_id, name, email/phone, role, credentials
Patient	Patient profile	patient_id, profile, preferences, caregiver linkage
Caregiver	Caregiver profile	caregiver_id, contact details, patient permissions
Person	Familiar-person registry	person_id, name, relationship, notes
FaceEmbedding	Recognition representation	person_id, embedding, model/version, metadata
Object	Tracked object type	object_id, name/category
ObjectObservation	Last-seen memory	object_id, timestamp, location/context, confidence
Conversation	Conversation record	conversation_id, participants,

		timestamp, transcript/metadata
MemoryEmbedding	Vectorized memory	memory_id, embedding, source, metadata
Medication	Medication master record	name, dosage/instructions, patient_id
MedicationSchedule	Medication timing	medication_id, time, recurrence, active period
Appointment	Calendar event	title, date/time, location, notes
Reminder	Notification rule	type, target, time, recurrence, status
DailySchedule	Routine item	activity, time, recurrence, status
Geofence	Safe zone	center, radius, name, active flag
Location	Location event	latitude/longitude or coarse location, timestamp
Alert	Caregiver notification	type, severity, timestamp, status
ActivityLog	Audit/activity history	event type, actor, timestamp, metadata

8. Constraints

- The project is intended as a college software project and must remain implementable within the available development time and hardware.
- AI performance will depend on the smartphone camera, lighting, microphone quality, model choice and available compute.
- Continuous camera/audio/GPS processing can have battery and privacy implications.
- Geofencing accuracy depends on device location services and environmental conditions.
- RAG quality depends on the quality and amount of stored conversation/event information.
- External cloud AI services should not be treated as mandatory because the project material emphasizes privacy-preserving/local processing.

9. Acceptance Baseline

The project is considered functionally complete for the baseline when the following end-to-end scenarios can be demonstrated:

1. Caregiver registers a patient, adds a medication and schedule, and the patient receives the reminder.
2. Patient asks for an object and the system returns its latest detected location with timestamp.
3. Caregiver registers a familiar person and the patient app recognizes the person from camera input under supported conditions.
4. A conversation is stored, later queried, retrieved through the RAG pipeline, and used to generate a grounded response.
5. Caregiver creates a safe zone and receives an alert when a simulated/test patient location crosses the boundary.
6. Patient can use the assistant to retrieve schedule, reminder and memory information.

10. Requirement Traceability Principle

Every implementation module, API, database entity and test case should map back to one or more requirement IDs. Requirements should not be added to the implementation without updating this baseline.

11. Source Basis

This requirements baseline is based on the user's handwritten project feature list and the supplied 19-page project presentation. The presentation describes the problem, six objectives, object-tracking pipeline, conversational RAG pipeline, face recognition/geofencing pipeline, data-flow architecture, technology stack, privacy-first architecture and future scope.